

Introduction to Web Technologies — Assignment 2

CSS fundamentals: selectors & priority, colors, fonts, spacing, alignment, classes & IDs · CSS layouts: Flexbox, Grid, positioning, float & clear, centering

Step 0. This is a continuation

You now style the pages you built in Assignment 1. **Same team, same theme, same repository — you do not start a new project and you do not create new pages.** The theme is now locked: you keep working on this same organization until the end of the course, and it cannot be swapped for a different one because it turned out to be inconvenient. If your organization has stopped answering you, come and tell me instead of quietly changing it. If your HTML from Assignment 1 was weak, fix it first: bad markup makes CSS painful, and that is the lesson of this week. The site still runs from local files; no domain, no hosting.

What to build

A css folder with two kinds of file. base.css is written by the team together and holds the colour palette, typography, and the styling of header, nav, main and footer that is shared by every page. Each student also writes their own stylesheet for their own two pages, named after them. Every page links base.css first and the personal stylesheet second, **so that you see the cascade and load order working.** A student working alone writes base.css plus one personal stylesheet.

Selectors — use each at least once

Type selector, class, id, descendant, child (>), adjacent sibling (+), grouping with commas, an attribute selector, the universal selector, the pseudo-classes :hover and :focus, one of :first-child or :nth-child, and one of the pseudo-elements ::before or ::after. Comment each of the less obvious ones with what it matches on your page.

Classes and IDs

At least eight classes and at least two ids. A class is reused on several elements; an id appears exactly once per page — write a comment saying why an id was the right choice there. **Names must describe meaning, not appearance:** .price-card and .opening-hours are fine, .red, .big, .div2 and .block1 cost 2 points each.

Colors, fonts, spacing, alignment

At the top of base.css list your palette in a comment: five colours at most, with a one-line reason for each choice. Write colours in hex, in rgb or rgba, and use one named colour. Define two font families with a full fallback stack, and set font-size, font-weight, line-height and letter-spacing deliberately rather than leaving browser defaults. Show that you understand the box model: box-sizing, margin, padding and border used on purpose, plus one comment explaining a margin collapse you ran into. Align text and elements with text-align and with layout properties, never with non-breaking spaces or empty elements.

Priority and the cascade

Everything lives in external stylesheets except two deliberate demonstrations: **exactly one internal style block and exactly one inline style attribute**, each with a comment saying that it exists to show the cascade and what it overrides. You may use !important exactly once, and only with a comment justifying it. Every additional inline style or !important costs 5 points.

Flexbox

Your navigation is a flex row using justify-content, align-items and gap. At least one more block on each student's pages is a flex container with flex-wrap and items that grow or shrink — a row of cards, services, prices or team members. Use flex-direction on at least one of them.

Grid

At least one section on each student's pages is a CSS grid using grid-template-columns with fr units, repeat() and gap, plus one item that spans more than one column or row. Use minmax() at least once. Add a comment saying why grid suited this block better than flexbox.

Positioning, float and clear

Demonstrate all four positioning values with a real purpose: static explained in a comment, relative as a containing block, absolute for something placed inside it such as a badge or a caption, and fixed for an element that stays on screen. Also float an image inside a paragraph of your own text and clear it correctly, then explain in a comment what would happen without the clear.

Centering

Centre three different elements using three different techniques — margin auto, flex, grid, or absolute positioning with transform. Label each one in a comment with the technique used and why it fits that case.

Forbidden

No CSS frameworks or libraries — no Bootstrap, Tailwind, Bulma or any other. No templates, no downloaded or copied stylesheets, no site builders, no JavaScript. Do not add wrapper divs to your HTML just to make a layout work: fix the CSS instead. Media queries and responsive design come later; design for a normal desktop window.

CSS checklist

Submit a checklist file naming every required selector, property and technique together with the stylesheet and the line number where you used it, and which student wrote it. Missing or wrong line numbers mean the item is not counted.

Specificity experiment — in your CSS, not in a report

Somewhere in your own stylesheets, make two rules target the same element on purpose so that they conflict. **Leave both rules in the file** and write above each one a comment giving its specificity value and which of the two wins. Then resolve the conflict without using !important. There is nothing to hand in for this beyond the stylesheet itself — no report and no screenshots — but you will be asked to walk me through it at the defense.

Design sketch

Before you write any CSS, draw the layout of two of your pages by hand on paper — signed, dated and photographed. Submit that photo together with a screenshot of the page before styling and after styling. Coding first and drawing afterwards is obvious and costs the full sketch score.

Submission

The same Git repository, extended. It must contain your stylesheets, the CSS checklist, the hand-drawn sketches, the before-and-after screenshots, an updated AI log and an updated README. There is no written report in this assignment. Everyone commits from their own account: at least four commits across three different days, six if you work alone. Your CSS must pass the W3C CSS validator with zero errors and your HTML must still pass the HTML validator.

Defense in class

I will ask you to explain a random rule from your stylesheet, to say what a selector matches, to walk me through your specificity experiment, to justify a choice between flexbox and grid, to say what would break if position: relative were removed from one of your parents, and to make a small live edit such as changing the number of grid columns or centring an element a different way. Your grade is the project score multiplied by how well you explain your own work; if you cannot explain it, the score is zero. **Work that is submitted but not defended is not evaluated, even if it was uploaded before the deadline.** The deadline gives you the right to be graded; the defense gives you the grade.

Grading (100 points)

Selectors, classes and ids 16 · colours, fonts, spacing and alignment 14 · priority, the cascade and the specificity experiment 18 · flexbox 15 · grid 15 · positioning, float and centering 16 · sketch and process 6. Penalties: 15 for any framework or copied stylesheet, 5 per extra inline style or !important, 5 per meaningless class name, 5 per validator error, 10 for wrapper divs added only to serve the layout.

AI policy

You may ask AI to explain concepts. You may not use it to write your stylesheets or your text, and every question you ask must be recorded in the AI log. The parts of this assignment that carry the most points — your hand-drawn sketches, your own stylesheets, your commit history and the live defense — cannot be produced by AI.